

Forecasts on AI and Modern Mercantilism

Modern mercantilism suppresses price signals to secure strategic inputs. AI changes the economics of sensing and codified output. The seventeen forecasts below price a single claim from seventeen directions: the decade will favor systems that can exercise control without disabling error correction.

Part 1: One Claim, Priced Seventeen Ways

Each probability begins with a reference class or market-implied prior, measures the distance between the current state and the threshold, and then adjusts for named drivers. Part 3 makes that reasoning explicit, including one derivation worked from premise to result. Table 2 specifies the resolution mechanics: sources of record, vintage rules, tie-breaks, and fallbacks.

- 1. 85%: More than 60.0% of the passenger vehicles China sells at retail in calendar 2027 are new-energy vehicles, resolving YES if the CPCA's first full-year 2027 print places the NEV share of retail sales strictly above 60.0%.**
- 2. 80%: The published on-demand output-token price of the most capable generally available model from at least one of OpenAI, Anthropic, or Google ends 2027 at least 50% below that provider's July 31, 2026 level, using dated pricing-page readings and the pre-registered selection rule in Table 2.**
- 3. 65%: China's average gasoline demand for calendar 2027 prints below its calendar-2026 average in the first IEA Oil Market Report carrying a complete 2027 figure.**
- 4. 70%: MP Materials remains named on China's MOFCOM export-control list on December 31, 2027.**
- 5. 68%: China accounts for more than 54.0% of global industrial robot installations in calendar 2027, as reported in the first print of the IFR's World Robotics 2028 edition.**
- 6. 35%: Before the end of 2028, an EU or member-state authority opens a formal proceeding against at least one of Meta, Microsoft, Alphabet, Amazon, OpenAI, or Anthropic specifically concerning employee-monitoring data used for AI training.**
- 7. 56%: Before the end of 2028, at least one of Microsoft, Alphabet, Amazon, Meta, or Oracle publicly discloses that it has cancelled or indefinitely deferred at least 1 GW of previously announced US datacenter capacity.**
- 8. 30%: Combined calendar-2028 purchases of property and equipment by Microsoft, Alphabet, Amazon, and Meta print below the combined calendar-2027 figure in first-print SEC filings.**
- 9. 30%: Before the end of 2030, a Fortune 500 company outside the Technology and Media sectors announces an executed agreement that licenses its employees' workflow data (screen recordings, keystroke-level logs, or software-interaction traces) for model training to OpenAI, Anthropic, Google, Meta, xAI, or Microsoft.**
- 10. 20%: A model from a China-headquartered lab occupies the #1 position on the LMArena overall text leaderboard for at least 30 consecutive days before the end of 2027, according to daily archived snapshots.**
- 11. 30%: At least two of Microsoft, Alphabet, Meta, and Oracle disclose a reduction in the stated useful life of server or network equipment in annual reports filed before March 31, 2029.**
- 12. 38%: Combined US goods imports from Vietnam and Thailand in calendar 2028 print below their combined calendar-2026 level in the Census Bureau's first-print FT-900 annual data.**
- 13. 55%: US core PCE inflation prints at or above 3.0% year over year in at least six of the twelve months of calendar 2027, using BEA first prints.**

14. 25%: Net customs duties collected by the US government in fiscal year 2027 exceed \$250 billion in the September 2027 Monthly Treasury Statement, first print.

15. 40%: Before the end of 2028, a Fortune 500 company outside the Technology sector publicly discloses, in a securities filing, official earnings-call transcript, or press release, that synthetic training data generated by a world model or physical-AI foundation model was used to train robots operating in its own production or logistics facilities.

16. 70%: Fewer than 20 of the 28 projects in LA Metro's Twenty-eight by '28 program are open for passenger revenue service on July 14, 2028, the opening date of the LA28 Games.

17. 30%: Between August 3, 2026 and December 31, 2027, USD/JPY moves at least 10% peak-to-trough within a single window of 20 consecutive trading days, in either direction, on the Federal Reserve's H.10 daily series.

Part 2: Framework

Modern mercantilism exchanges the price mechanism for command over strategic inputs. AI sharply reduces the cost of sensing and codified output. The decisive question is whether institutions can exercise control while preserving the capacity to detect and correct error.

Modern mercantilism treats trade and technology as instruments of state power. Governments override market prices through tariffs, subsidies, export controls, and directed procurement to secure inputs they consider strategic. Every institution depends on a feedback loop: observe conditions, retain what happened, form a prediction, act, and register the result. Mercantilist policy weakens the loop's clearest signal, relative prices, to gain control over supply. AI changes the first stage of that loop by making telemetry and measurement far less expensive, while rendering some older signals obsolete.

The outcomes diverge according to whether error correction survives the suppression of prices. We define that capacity operationally. When a delivery or demand signal moves against a standing commitment, a system preserves error correction if some recorded institutional behavior, such as supply, accounting recognition, docketed enforcement, or a published revision, adjusts in the same direction within roughly twenty-four months. Several forecasts measure this elasticity directly. Forecast 11 asks whether depreciation schedules respond to hardware turnover within the window. Forecasts 7 and 8 ask whether capital expenditure responds to the grid, while forecast 3 tests whether a demand response persists after the price shock that produced it has normalized. Successful systems direct capital while preserving contestation. Failed systems acquire command and gradually lose the means to discover their own errors.

The loop, and the six relationships it produces

Mercantilism and AI are the subject of this paper, but the mechanism is general. Societies operate as nested feedback systems, and the same chain runs at every level from a household to a state: local conditions shape perception and memory, perception feeds prediction, prediction drives action, action produces economic and policy feedback, and that feedback either becomes learning or accumulates as unrecognized error until it forces structural change. Every stage of that chain runs through people rather than around them, and it inherits the dispositions people actually have: memory weights the recent and the vivid, and prediction defends whatever the predictor has already committed to in public. Mercantilist policy and cheap machine sensing both intervene on this chain, at different stages. Tariffs, subsidies, and export controls degrade the feedback signal. AI reprices perception and prediction. The interaction is what the forecasts below measure.

Six relationships recur across domains, and each is priced by named forecasts rather than asserted. The table states what would have to happen for each to be wrong.

Relationship	Mechanism	Priced by	Falsified if
Information quality drives institutional performance	Elections, opposition, independent media, local government, and transparent measurement expose error; hierarchy, fear, and propaganda suppress bad news until mistakes compound	6, 7, 11	A visible delivery gap persists while no filing, docket, or published revision records it
Bounded rationality can destabilize markets	Emotion, imitation, and optimistic expectations produce correlated positioning; prices validate the expectation until leverage turns a reversal into a cascade	2, 8, 17	Crowded positioning and rising leverage unwind smoothly, with no disorderly move
Inequality feeds back into political power	Concentrated wealth buys influence over rules, media, and regulation; favorable rules deepen concentration; falling perceived fairness produces populist reaction	6, 9	Workflow data becomes broadly licensed and employment law fails to bind its accumulation
Policy delay compounds harm	Late intervention widens the gap it was meant to close, in learning, health, climate adjustment, and infrastructure alike	7, 8, 11, 16	Hard-deadline programs land on time without cost escalation or scope reduction
Conflict is a coupled prediction loop	Historical memory produces hostile expectations; defensive acts read as offensive to rivals; each response confirms the other's forecast, with verification and intermediaries as the damping term	4, 5, 10, 12, 15	Export-control listings lapse or controls relax without any verification mechanism appearing
Persistent mismatch can trigger nonlinear change	Inequality, polarization, leverage, environmental stress, and military buildup accumulate without immediate collapse; a trigger then reorganizes norms, institutions, and incentives at once	8, 13, 14, 17	Every accumulating mismatch we track resolves through smooth adjustment

Underneath several of these sits a mechanism that is human before it is institutional. A feedback loop closes only when some particular person acts on the signal, and acting early on an adverse signal is privately expensive. An executive who cancels announced capacity converts defensible optimism into a documented misjudgment. A finance officer who shortens a depreciation schedule concedes in a filing that competitors read that the previous estimate was wrong. The individual absorbs a concentrated cost so the institution can capture a diffuse one, so the private incentive is to wait until the evidence is undeniable and the decision can no longer be blamed on anyone's judgment.

The information usually arrives on time. What arrives late is someone willing to be the first to say so. That is why relationships 1, 4, and 6 produce the same observable signature: the correction happens, and it happens later and more sharply than the underlying conditions warranted. The lag is the forecastable part, and it is what forecasts 7, 11, and 16 price. Forecast 7 splits it in two explicitly, separating the decision from the admission.

The sixth relationship needs a caveat, because it is the one most easily abused. A social phase transition, the symmetry breaking that follows accumulated mismatch, has no resolution criterion we can write for a three-year window: reorganization is recognizable afterward and unfalsifiable in advance. So we do not forecast one. What we forecast are the accumulating mismatches that would precede it, each with a threshold and a source of record, and we treat their joint resolution as evidence about pressure rather than about timing. A reader who wants the strong claim tested should watch forecasts 8, 13, and 17 together: a capital-expenditure reversal, an inflation floor that holds, and a disorderly funding move are the observable signature of pressure discharging rather than dissipating.

The third relationship carries the thinnest coverage in our set, and we would rather say so than imply otherwise. Inequality and political power enter this paper only through who ends up owning workflow data and whether

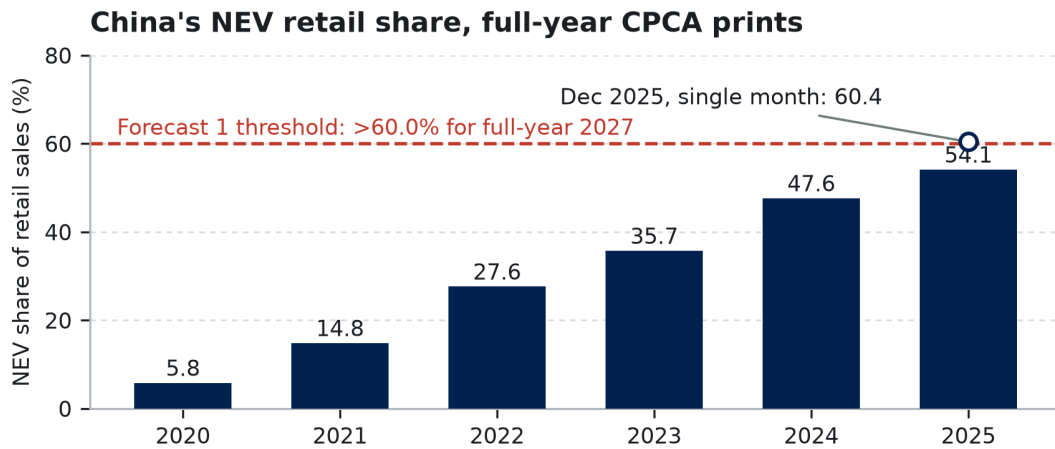
employment law constrains its accumulation. We price no forecast on wealth concentration, campaign finance, or regulatory capture directly. That is a real gap between the framework's breadth and the evidence we put at risk.

The five claims that follow apply this chain to mercantilism and AI specifically.

Framework claim	Priced by	Where we differ most from consensus
Strategic control pays during crises, as optionality	1, 3, 4, 12, 14	Post-shock consensus expects oil demand to recover at lower prices; we expect China's gasoline line to decline (3)
Physical delivery becomes the binding constraint	7, 8, 11, 16, 17	We found no published model of a 2028 hyperscaler capex contraction; we assign 30% (8)
Workflow data is the chokepoint; employment law sets accumulation	6, 9	Data marketplaces are repeatedly predicted; we assign 70% to the moat outcome (9)
As codified knowledge depreciates, scarce inputs gain value	2, 5, 10, 15	China loses the frontier scoreboard while winning diffusion (10 and 5 as a pair)
The chain produces an inflation floor	13, 14	The June 2026 SEP raised 2026 core PCE to 3.3% and penciled a hike; we expect persistence through 2027 (13)

1. The return on mercantilism appears during crises, in the form of optionality. Conventional measures such as market share and trade balances miss much of the payoff. Strategic control remains latent in ordinary periods and becomes valuable when supply is disrupted. China's EV program supplies the clearest example. A campaign for automotive competitiveness became a form of energy security. By the time the Hormuz disruption arrived this spring, China's electric fleet had displaced roughly one million barrels a day of oil demand (Kpler estimates approximately 540 kb/d of gasoline and 500 kb/d of diesel in 2026). That reduction allowed China to withdraw about four million barrels a day from normal purchasing. On IEA accounting, crude imports fell roughly 40%, refineries reduced runs, and product exports were curtailed as Brent rose above \$113 and then returned below \$70. Demand elasticity absorbed part of the shock, supported by capacity built years earlier at subsidized cost. Forecasts 1 and 3 extend the argument: the transition continues with purchase incentives at half their former cap, CPCA monthly prints have remained above 60% since April 2026 (61.4% in April, a record 63% in May, 62.8% in June), and the oil-demand break persists through the post-shock rebound.

One correction to our own reading. A share is a ratio, and we had been reading it as adoption. NEV retail units fell 5% year over year in May and 7% in June while total passenger-vehicle sales fell 20% in April, so penetration rose because the denominator contracted faster than the numerator. Part of what Figure 1 shows is internal combustion collapsing inside a shrinking market. A competing account credits the halved purchase-tax exemption with depressing NEV volumes for reasons unrelated to the ICE decline, and published retail data cannot separate the two. Forecast 1 survives this better than our mechanism does: a share driven by denominator collapse is thinner evidence that strategic control pays than one driven by adoption.

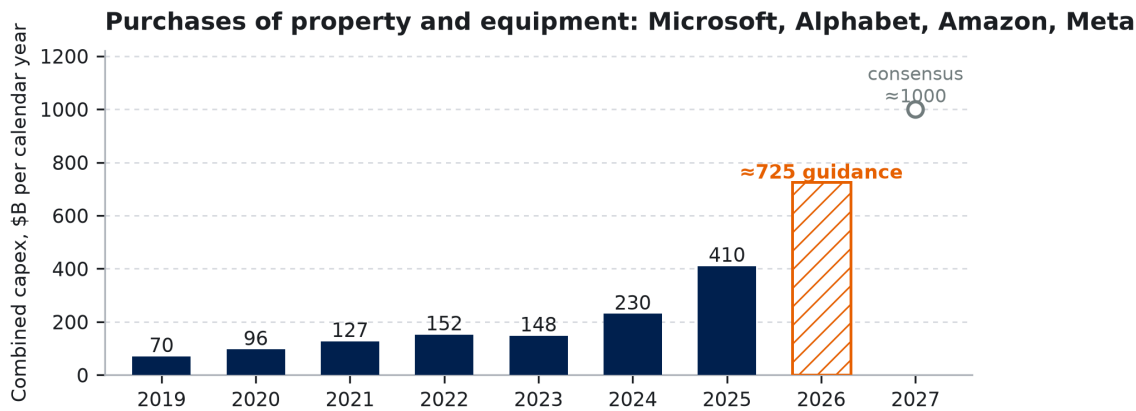


Source: China Passenger Car Association retail data (BEV+PHEV units ÷ total passenger-vehicle retail units), full-year first prints 2020–2025 and the Dec 2025 monthly print (hollow marker). [data/fig01_nev_share.csv](#) lists each print and URL.

Figure 1. CPCA retail data, full-year prints. The share reached 54.1% in 2025 with purchase-tax relief at half its former cap, then crossed 60% in a single month for the first time in December. Forecast 1 asks the full year to clear that threshold by 2027.

The comparison cases reveal weaker correction. ASEAN's China+1 expansion relied on a tariff arbitrage now compressed by the July Section 301 action (forecast 12). Thailand, with a fraction of Japan's income per person, placed battery-electric vehicles at 19.4% of 2025 sales (FTI data), mostly Chinese brands. Japan, several times richer, remained below 3% even including plug-in hybrids. Income predicts the reverse ordering. Policy-shaped access predicts the observed one, and a larger subsidy preceded Japan's adoption jump in early 2026. Availability at a given price point determines what people drive, and availability is set upstream by trade and subsidy policy. Preference adjusts to what reaches the lot.

2. The binding constraint has moved from capital to physical delivery. Capital is abundant; transformers, interconnection capacity, skilled trades, and permitting throughput are scarce. Roughly \$725 billion of calendar-2026 capital expenditure guided by Microsoft, Alphabet, Amazon, and Meta depends on the grid, up about 77% on 2025's \$410 billion, with analyst consensus now above \$1 trillion for 2027. Divergence between announced and energized capacity makes the gap forecastable. Forecast 7 prices a gigawatt-scale cancellation; forecast 8, a spending decline; forecast 11, a shorter server-depreciation schedule that concedes the original assumptions failed. One layer below, Beijing's export controls influence who can manufacture the permanent magnets in every humanoid actuator (forecast 4).



Sources: company SEC cash-flow statements, calendar-quarter sums of “purchases of property and equipment,” finance leases excluded (navy = reported actuals). Hatched 2026 = company guidance, on boundaries that differ between companies; hollow 2027 = analyst consensus, a projection and not guidance. Three epistemic classes, three marks. data/fig02_capex.csv.

Figure 2. Purchases of property and equipment, calendar years. Actuals run through 2025; the 2026 figure is guidance measured on a boundary that is not strictly comparable. The series' one previous decline occurred from 2022 to 2023. Forecast 8 asks for a second at ten times the scale.

3. Workflow data is the new chokepoint, and employment law determines how quickly it accumulates. Frontier AI increasingly depends on proprietary workflow data: records of how people actually operate software. These data are largely absent from the open internet. They arise inside firms through employee activity, so their collection is governed by labor and privacy law, with consequences that extend into trade. Meta's Model Capability Initiative offers an early test case. The program captured screen content and keystroke-level activity from US employees before a security review paused it in June. Meta has confirmed the program, the lack of a broad opt-out, and the pause; its CTO stated there is no opt-out, and more than 1,600 employees signed an internal petition demanding cancellation. Press reporting based on an internal security notice placed the exposed corpus at roughly 45,000 internal tables, a figure Meta has not confirmed. The detail that carries the most weight for this claim is jurisdictional: European employees were excluded from the program because GDPR does not permit the collection. One firm, one policy, one program, running in one jurisdiction and barred in another. We had argued this claim from statute; the carve-out argues it from behaviour. The pause leaves both the corpus and the incentive intact. US law generally permits such accumulation. In the EU, GDPR minimization requirements, national workplace-monitoring statutes, and works-council vetoes impose much tighter limits (Thread 2). Employment regulation therefore sets the accumulation rate of one of the decade's scarcest inputs. Europe may protect workers while excluding its firms from the underlying input market. We expect enforcement to emerge (forecast 6) and the data to remain concentrated inside firms as a moat instead of becoming a widely traded commodity (forecast 9).

4. As codified knowledge becomes less expensive, value shifts toward persistent scarcities. The gains move toward tacit skill and physical position. The change first appears in the price of model output (forecast 2, Figure 5), perhaps the most fully codified product in the economy. It also makes inference demand less durable than current capital expenditure assumes, because interchangeable models invite workloads to migrate toward the lowest acceptable price. Where automation spreads is the clearest case, because the standard account gets the geography backwards. Automation is normally explained by the price of labor: machines arrive where workers are expensive. The installations run the other way. China has taken most of the world's industrial robots every year since 2021 while its manufacturing wages remain a fraction of US levels and its working-age population shrinks. Japan, whose working-age population has contracted for more than a decade, has pushed self-checkout through the large majority of its supermarkets (the Japan Supermarket Association's survey is the source, carried from earlier reporting and not re-verified in this revision). Amazon's checkout-free format in the high-wage US market leaned on offshore human reviewers and came out of full-size US grocery stores in 2024.

Scarcity is doing the work that cost usually gets credited with. Where labor is abundant, firms keep people in the process because people are available and adaptable, and the automation decision can be deferred indefinitely without

anyone deciding anything. Where labor genuinely cannot be hired, a staff-less format stops being an efficiency project and becomes the condition for staying open. That distinction predicts a geography the labor-cost account cannot: robots concentrating in a middle-income manufacturing economy with a shrinking workforce, while the richest markets keep people in the loop. China's own position then splits deliberately. Compute export controls keep its models off the frontier leaderboard (forecast 10) while low barriers to deployment push automation through physical industry (forecast 5).

5. The macroeconomic block follows from the same chain. Trade instruments alter relative prices. Those changes redirect investment toward strategic capacity and away from measured efficiency. The resulting investment mix determines the balance between growth and inflation, which in turn constrains fiscal and monetary policy. Forecasts 13 and 14 read two gauges along that sequence: the inflation floor created by the policy wedge and the tariff regime's own revenue line, the clearest observable measure of whether the wall remains in place.

How we would know we are wrong. The framework rises or falls on patterns across forecasts. If capital expenditure compounds through 2028 while energized capacity keeps pace (7, 8, and 11 all NO), the claimed delivery constraint disappears. If Chinese gasoline demand rebounds (3 NO) as the EV share stalls (1 NO), the optionality account of mercantilism fails in its strongest case. If a Chinese lab sustains the frontier lead while industrial diffusion stalls (10 YES and 5 NO), control and competence move together, erasing our central distinction. A liquid and broadly accessible workflow-data market by 2028 (9 YES) would defeat the chokepoint thesis. Core PCE below 3.0% for most of 2027 (13 NO) would sever the proposed transmission from policy wedge to prices at its first observable link. If the LA28 program delivers 20 or more of its 28 projects on the opening date (16 NO), the delay relationship fails on the one hard deadline we test outside the AI buildout, where it should bind most visibly. Any three of these patterns would be fatal. The forecast set gives the framework a genuine route to failure.

One asymmetry is worth stating plainly, because it limits what our low-probability forecasts can prove. Forecast 17 at 30% cannot falsify anything by resolving NO: an orderly seventeen months is what we already expect. Its informative direction is YES, which would supply the cascade relationship its clearest confirmation. The same holds for forecast 15 at 40%. Only the forecasts we price above even odds carry falsifying power in the NO direction, and readers should weight them accordingly.

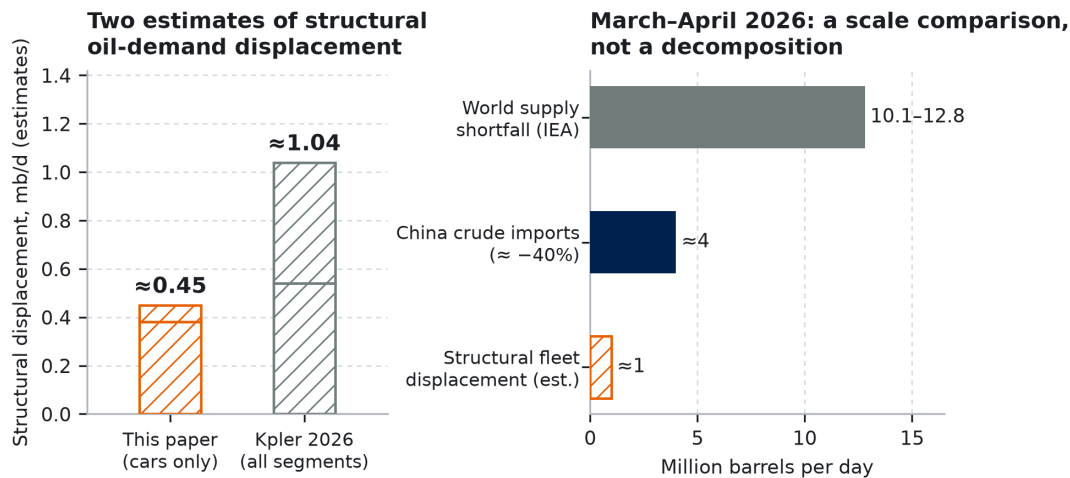
Part 3: Analytical Appendix

Derivations in which arithmetic carries the argument, followed by resolution mechanics and a self-audit of the claims most likely to fail.

Thread 1: China's Oil Position Is Arithmetic, and the Crisis Tested It

The displacement estimate is arithmetic. China's NEV fleet reached 31.4 million vehicles at the end of 2024 (Ministry of Public Security registrations), 70.3% of them battery-electric. At 13,000 km of annual travel and a counterfactual internal-combustion efficiency of 7.5 L/100km, battery-electric vehicles displace 21 to 22 billion litres of gasoline a year. Plug-in hybrids at half electric mileage add roughly 4 billion litres. Passenger vehicles therefore displace approximately 0.45 million barrels a day, an estimate identified in Figure 3. Kpler independently reaches roughly 1.0 mb/d across all electrified segments in 2026: about 540 kb/d of gasoline and 500 kb/d of diesel.

The crisis mechanics matter more than the level. On the IEA's monthly accounting, world supply fell roughly 10 to 13 million barrels a day below pre-conflict levels: 10.1 mb/d in March and 12.8 mb/d in April. China reduced crude imports about 40%, withdrawing roughly four million barrels a day from normal purchasing, while refineries cut runs and product exports. Structural fleet displacement supplied the smaller share of the adjustment; refinery cuts, curtailed exports, and inventory draws supplied most of the volume. These measures avoided domestic shortage because baseline gasoline demand had already fallen. The fleet made the larger adjustment survivable. Subsidies purchased that option years earlier, while trade-balance measures recorded only its cost.



Sources: derivation from MPS fleet registrations and stated assumptions (Thread 1), BEV 0.38 + PHEV 0.07; Kpler displacement estimates, 2026, gasoline 0.54 + diesel 0.50; IEA Oil Market Report monthly accounting, Mar–Apr 2026. Hatched bars are estimates. The three right-hand quantities are different kinds of measure, shown together only for scale; refinery-run cuts and halted product exports, not the fleet, supplied most of the import swing.

Figure 3. Left: two independent estimates of structural displacement, both identified as estimates. Right: a scale comparison among the supply shortfall, China's import reduction, and the structural displacement that made the reduction survivable. The panel does not present a decomposition.

Forecast 3 turns the argument into a ratchet test: does the decline persist after the price pressure recedes? Gasoline demand is tracking a roughly 5.5% decline this year (GL Consulting via Bloomberg, versus its 5.2% prewar estimate), a fall the IEA notes would be the second steepest on record after the 2022 lockdown year, while post-shock outlooks anticipate recovery as prices normalize. Any year-over-year rebound resolves NO, keeping the forecast at 65%.

The IEA proposes a different mechanism for the same decline, and we name it rather than let a reader find it. Its reading is that high pump prices discouraged internal-combustion driving, with electric vehicles compounding the effect. Ours is that a fleet built years earlier removed baseline demand permanently. The two predict the same 2026 and different 2027s, which is what forecast 3 adjudicates: price-induced destruction should partly reverse once Brent settles, and fleet displacement should not. Second-quarter crude imports of 8.1 mb/d, 32% below the prior quarter and the first sub-8.0 months since 2016, dwarf either fleet estimate in Figure 3, consistent with our position that refinery cuts and curtailed exports supplied most of the volume. Weaker correction produces the ASEAN fork. The July 23 Section 301 action places sixty economies at 10% or 12.5%, with Vietnam and Thailand at the higher tier and specified carve-outs (USTR), narrowing the arbitrage that financed China+1. Forecast 12 assigns 38% to the tariff wall overcoming relocation momentum that survived every previous round. If forecast 13 resolves NO while forecast 6 resolves YES, capacity automated in place.

Thread 2: Workflow Data Is a Chokepoint, and Employment Law Is the Trade Policy

The legal asymmetry is already observable. In the United States, employers may generally monitor company equipment under the business-purpose exception to federal wiretap law, subject to notice laws in several states. The European Union constrains collection through GDPR minimization and purpose limitation, Italy's Workers' Statute Article 4 and its counterparts, and works-council codetermination over monitoring systems. The AI Act adds little before December 2027: the Digital Omnibus (Regulation (EU) 2026/1744) deferred high-risk obligations for workplace AI from August 2026 to December 2, 2027. Although written for other purposes, these rules act as industrial policy by controlling the accumulation of human software-use demonstrations, which agentic models increasingly require and public sources lack. Meta's MCI shows accumulation where law permits it. The pause interrupted collection while leaving the corpus and incentive intact.

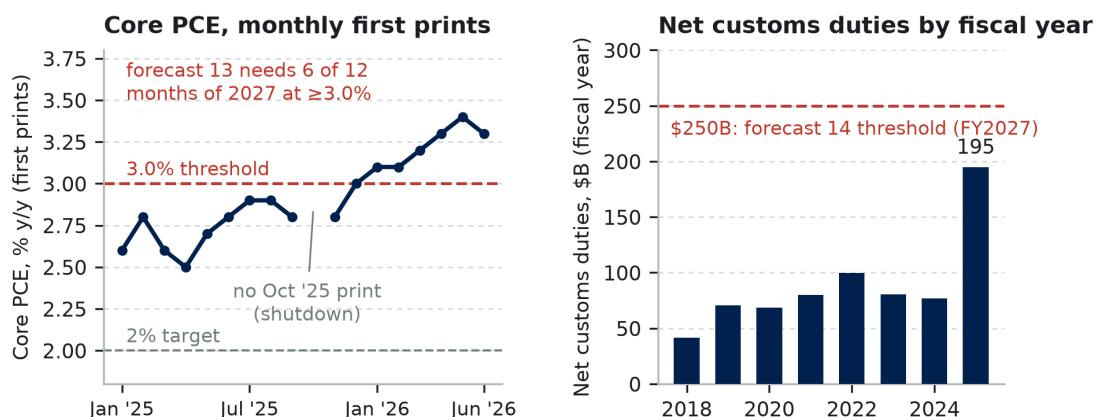
Forecast 6 prices the enforcement side of this collision, where the base rate is thin. Across eight years of GDPR enforcement, exactly one formal proceeding against the six named firms has concerned employee-monitoring data:

CNIL's case against Amazon France Logistique, decided in December 2023 and followed by a reduction of the fine on appeal. Exactly one has concerned AI-training data: the Garante's action against OpenAI, annulled on jurisdictional grounds in March 2026. No authority has combined both elements in a single docket. The closest analogue is the Irish DPC's 2025 inquiry into X over Grok training data, involving a firm outside the specified six. Several considerations raise the probability above that sparse record. Employee-data cases proceed through national authorities rather than the Irish bottleneck, the labor channel can move quickly once activated, and the MCI episode increased the salience of this precise conjunction. The prior draft assigned 62% based on an AI Act trigger absent from the first sixteen months of the window. A fresh calculation gives 35%. The forecast requires three events: a visible trigger, an authority willing to name the AI-training nexus in its scope, and a formal opening before the window closes. The high-risk regime widens that channel only during the final thirteen months.

Forecast 9, the moat-versus-market question, assigns 30% to the repeatedly predicted emergence of data marketplaces. Three frictions obstruct that market. Arrow's information problem comes first: a buyer cannot value workflow data without inspecting it, yet inspection transfers part of the value. Adverse selection follows, because a firm's willingness to sell its workflow exhaust may signal that the data contain little distinctive knowledge. Liability compounds both problems, since employee personal data cannot be separated cleanly from the underlying record. Strategic retention completes the argument. An early sign of error would be the appearance of brokers that certify and anonymize corpora, as clinical-claims aggregators did for healthcare data. A broker-mediated market would still resolve forecast 9 NO under the stated criterion, so the thesis can fail before the score records that failure.

Thread 3: The Macro Block, Derived from First Prints

All macroeconomic anchors are first prints dated July 30, 2026. Core PCE reached 3.3% year over year in June, its seventh consecutive reading at or above 3.0% (BEA). The 3.50 to 3.75% policy rate continued for a fifth meeting after a 9 to 3 vote, with three dissents favoring an increase (FOMC, July 29). The 30-year Treasury yield exceeded 5.2%, its highest since 2007 (Federal Reserve H.15), while second-quarter real GDP grew at a 1.5% annualized rate (BEA advance). Net customs duties totaled \$194.9 billion in FY2025. The tariff regime that produced that figure no longer exists in the form that produced it. On February 20, 2026 the Supreme Court held 6 to 3 that the International Emergency Economic Powers Act does not authorize tariffs, the taxing power being reserved to Congress, invalidating the reciprocal and the China, Canada, and Mexico tariffs from inception rather than prospectively. The Court of International Trade ordered roughly \$165 billion refunded; the Penn Wharton Budget Model put revenue at risk above \$175 billion. Customs and Border Protection opened phase one of refunds on April 20, 2026. FY2026 collections through June ran \$163 billion, and June itself printed **negative \$25.6 billion** as \$49.2 billion of refunds outran \$23.6 billion of gross duties (Treasury MTS).



Sources: BEA Personal Income and Outlays, monthly first prints, Jan 2025–Jun 2026 (October 2025 has no print, shutdown-delayed reporting; drawn as a gap, not interpolated); US Treasury Monthly Treasury Statements, net customs duties by fiscal year, complete fiscal years only; the partial FY2026 total is excluded rather than annualized. Rows, statuses, and URLs: data/fig04a-b CSVs.

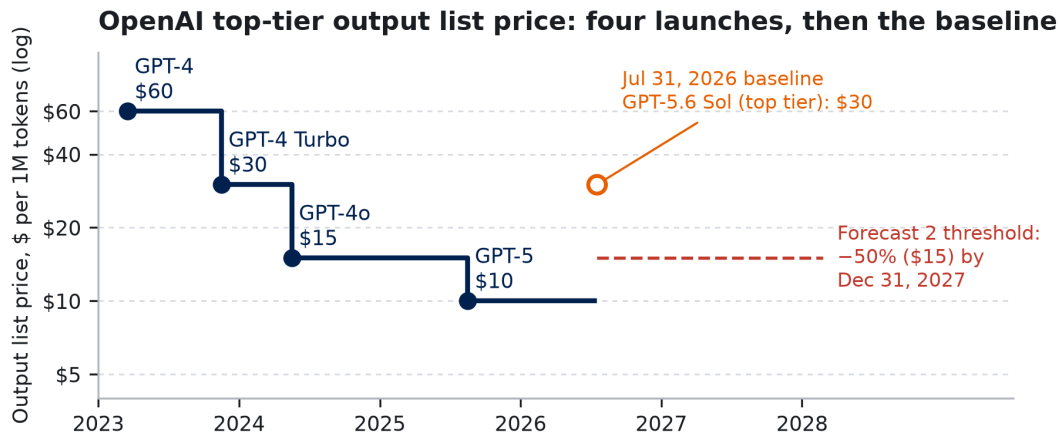
Figure 4. Two gauges at the July 2026 anchors, first prints throughout. Every 2026 core PCE print sits at or above 3.0%, including the 3.3% June reading taken after the tariff wedge began unwinding. FY2025 duties rose \$118 billion in one year to \$194.9 billion; FY2026 then broke, with the Supreme Court's February 20 ruling invalidating the IEEPA tariffs and June net collections printing negative \$25.6 billion. The gap to forecast 14's threshold is no longer a growth question.

Forecast 13, worked end to end. Since 1985, the revised core-PCE series has moved durably from above 3.0% to below it only twice: during the early-1990s recession and in late 2023, when supply shocks unwound. Neither episode coincided with trade policy actively adding costs. Decompose the event: $P(\text{YES}) = P(\text{core enters 2027 at or above 3.0\%}) \times P(\text{six qualifying months given that start}) + P(\text{enters below}) \times P(\text{six qualifying months anyway})$. We assign 0.65 to the first condition. Seven consecutive prints have met or exceeded 3.0%, the latest three reached at least 3.3%, and the July Section 301 round continues to pass through despite restrictive monetary policy. Near-term consensus has moved in the same direction: the June 2026 SEP raised projected 2026 core PCE to 3.3% and shifted its median dot to an increase. Our disagreement concerns persistence through 2027. Conditional on beginning the year at or above the threshold, we assign 0.75 to six qualifying months. Beginning below it, we assign 0.15 to a six-month re-acceleration. The result is $0.65 \times 0.75 + 0.35 \times 0.15 \approx 0.54$, reported as 55%, a modest lean toward persistence. Because inflation from any source resolves YES, forecast 13 establishes co-movement with the framework rather than causal attribution. Calibration.md records the inputs and sensitivity. **Forecast 14, repriced from 70% to 25%.** The prior draft treated the tariff wall as entrenched and asked only whether \$194.9 billion could grow by \$55 billion. That framing assumed the FY2025 regime persisted. It did not. Three things now work against the threshold at once. Refunds subtract from the same net line the forecast resolves on, and have already carried one month below zero. The replacement authorities are partial: the Committee for a Responsible Federal Budget puts Section 301 and 338 together at under 60% of the lost IEEPA revenue, and scores post-January-2025 tariff actions \$825 billion below CBO's February 2026 baseline through FY2036. The bridge authority is contested too, February's Section 122 tariffs having been ruled illegal by the Court of International Trade pending appeal. Clearing \$250 billion net in FY2027 requires a successor regime to beat the IEEPA-era gross run rate while the refund tail is still being paid. We hold 25% rather than less because Section 301 now covers roughly \$949 billion of imports at 10 to 12.5%, the refund tail should largely clear during FY2027, and Congress retains the power to ratify tariffs by statute. This is the one forecast in the set that a single external event repriced, and we would rather show the revision than bury it.

Shorter Derivations

Forecast 2 (token prices). OpenAI's flagship launch price fell from \$60 to \$30, then \$15 and \$10, across twenty-nine months; two steps exceeded 50% within seventeen months. The market's top tier has since repriced upward. On July 31, 2026, OpenAI's designated premium model, GPT-5.6 Sol, carried a \$30 output price, while July reductions applied to lower tiers (Figure 5). The forecast requires one of three providers to reduce its top-tier price by 50% from its own July 31, 2026 baseline within seventeen months. All three are likely to launch successors during that period. Earlier successors generally debuted at or below predecessor prices, although the latest did not. Three opportunities

and declining hardware costs compete with demonstrated premiumization. We assign 80%, down from 90%. The pre-registered selection rule resolves ambiguity around "most capable": the provider's pricing page determines its leading general-purpose tier on the baseline date. If the page makes no designation, the highest-priced generally available text model governs. Successors inherit the position; separately branded premium tiers count only when the page places them first.



Source: OpenAI published on-demand API list prices for the pricing page's top general-purpose tier; one provider, shown as the reference class; no quality adjustment. The hollow marker is the July 31, 2026 baseline the forecast resolves against, drawn as an estimate because direct pricing-page capture was blocked in this build (data/fig05_tokens.csv; see data/snapshots/README.md).

Figure 5. Four OpenAI top-tier launch prices from one provider, shown as the reference class, together with the July 31, 2026 baseline against which the forecast resolves. That baseline sits above the 2025 launch price. The dashed line marks the threshold at half the baseline.

Forecast 5 (robots). IFR World Robotics 2024 reports 276,288 Chinese installations out of 541,302 worldwide in 2023, or 51.0%. The 2025 edition reports 295,000 out of 542,076 in 2024, a computed 54.4% that the IFR's prose rounds to 54%. China's share rose from about 27% in 2015 to a majority in every year since 2021. The threshold requires the latest share to remain approximately flat. Our 68% probability allows for a meaningful interruption in Chinese capital expenditure while preserving confidence in the longer direction. If the IFR discontinues the series, the forecast remains unresolved. MIIT data for China cannot be paired with a stale global denominator.

Forecasts 7 and 8 (the delivery pair). Forecast 7 multiplies two probabilities derived separately. The first is a 0.80 probability that at least one of five companies cancels or indefinitely defers one gigawatt of announced US capacity by the end of 2028. Five companies, twenty-nine months, and announced pipelines that exceed any credible energization schedule justify that estimate. The second is a 0.70 probability that the company itself quantifies the decision on the record, conditional on its occurrence. Impairments and contract exits eventually require filings, and the criterion also accepts earnings-call transcripts. The most prominent pullback to date provides direct counterevidence. Analyst reports of Microsoft lease cancellations in 2025 and 2026 supplied capacity estimates and prompted a corporate clarification, yet the company never quantified the reduction. That second probability is where the human layer enters the arithmetic. Cancelling is an operating decision a company can take quietly. Quantifying it is a statement about the judgment that produced the original commitment, and it gets read as one. The 0.30 we withhold is the chance that the retreat happens and nobody attaches a number to it. Analyst reporting now puts that episode at roughly 1.5 GW of frozen self-build plus a gigawatt of lapsed agreements: past our threshold in substance, short of it in disclosure. The competing reading deserves weight, though. Microsoft leased \$11.1 billion of capacity in the first quarter of 2026 and Meta's CoreWeave commitments reached roughly \$35 billion, so capacity may be moving between owned and leased rather than leaving the system, and aggregate spending rose 77% in 2026. Forecast 7 asks only for a disclosed gigawatt, so both readings can hold at once. The product is 56%, replacing the draft's 65%, which treated disclosure and cancellation as the same event. Forecast 8 supplies the trailing confirmation: a year-over-year contraction in the four-company capital-expenditure total, assigned a 30% probability. We found no published sell-side estimate modeling such a contraction. The plotted history contains one decline

across six transitions, from 2022 to 2023, making 30% a generous prior. Conditional coherence with forecasts 7 and 11 nevertheless raises the estimate above what that distribution alone implies. The relevant reference class is the 2000 to 2002 telecom buildout, when US carrier capital expenditure fell by roughly half from its peak within three years.

Forecast 11 (depreciation). Recent history consists almost entirely of extensions, followed by the first reversal (Table 1). The effect on income is mechanical. For \$100 billion of gross server assets, reducing straight-line depreciation from six years to four adds roughly \$8 billion in annual expense. The criterion therefore requires two companies and excludes Amazon, which has already shortened its estimate. We assign only 30% because firms have a powerful incentive to delay recognition, and accounting rules permit that delay long after the underlying economics have changed.

Table 1. Stated useful lives of server and network equipment (company filings and earnings disclosures; the changes forecast 11 prices).

Company	Change	Effective	Equipment class	Old → new stated life
Microsoft	Extension	FY2023 (announced July 2022)	Servers and network equipment	4 → 6 years
Alphabet	Extension	January 2023	Servers; network equipment	4 → 6 years; 5 → 6 years
Amazon	Extension	January 2024	Servers	5 → 6 years
Meta	Extension	January 2025	Servers and network equipment	To 5.5 years
Amazon	Reduction	January 2025	Subset of servers and network equipment	6 → 5 years; plus a separate \$920 million Q4-2024 accelerated-depreciation charge on early-retired hardware

Forecast 10 (LMArena). Compute export controls bear most directly on the technological frontier. We expect China to lose this scoreboard while winning industrial diffusion (forecast 5); the contrast between the two outcomes is the substantive claim. The 20% probability allows for occasional #1 releases while recognizing the difficulty of remaining first for 30 consecutive days. If LMArena is discontinued, the forecast remains unresolved. A different benchmark could reverse the result and therefore cannot serve as a substitute.

Forecast 15 (world models). The prediction stage of the loop is being industrialized, and 2026 is the year it became a product rather than a demo. DeepMind's Genie 3 generates persistent interactive 3D environments in real time at 24 fps and ships to subscribers through Project Genie. NVIDIA's Cosmos 3, released in June, is an omnimodal world model aimed explicitly at physical AI, with embodied reasoning, task planning, and action modeling across vehicles, robots, and egocentric human motion; NVIDIA reports roughly two million cumulative Cosmos downloads. Benchmark attention has shifted from whether generated video looks real toward closed-loop usefulness, which is the honest measure.

Our framework predicts an asymmetry here, and forecast 15 prices it. World models are an attempt to codify the physical world, and by claim 4 the codified part gets cheap fast: fidelity and availability should improve quickly, and they have. What stays scarce is real interaction data and physical position, so the binding constraint moves to deployment rather than model quality. The forecast therefore asks not whether world models improve but whether an industrial operator outside technology puts one in its own robot training loop and says so on the record. Vendor adoption is already visible; end-user disclosure is the lagging indicator, exactly as in forecast 7. We price 40%, above forecast 9's 30% because vendors actively co-market customer deployments while workflow-data licensing carries legal exposure that suppresses disclosure, and below even odds because two years is short for a disclosed production deployment in a capital-intensive industry.

Forecast 16 (LA28 delivery). A fixed date the host cannot move is the cleanest available test of the delay relationship. LA Metro's Twenty-eight by '28 program is measured against the LA28 opening on July 14, 2028. The program list already carries its own evidence: a March 2024 board action replaced eleven of the original projects with

more achievable substitutes, which is a delivery constraint being recognized in public, and that revised list governs resolution. We price 70% that fewer than 20 of the 28 are carrying fare-paying passengers on the opening date. This is the delivery claim tested outside the AI buildout, which matters because forecasts 7, 8, and 11 all draw on one industry; if physical delivery is the binding constraint of the decade, it should bind a transit program on a hard deadline as well. An earlier draft cut this forecast for reaching the thesis only by analogy. The general form of the loop is what brings it back: relationship 4 is not about datacenters.

Forecast 17 (the yen). Relationship 2 predicts fragility rather than direction, and the yen is the cleanest priced instrument of it. Yen funding sits beneath a meaningful share of the leverage financing the AI buildout, so a disorderly move is one way the delivery constraint gets marked to market early. The reference event is the August 2024 carry unwind, roughly 12% in five weeks. We price 30% for a repeat inside seventeen months, roughly double the unconditional base rate.

Two honest qualifications. First, the criterion is deliberately blind to cause: a generic carry unwind resolves it YES with no mercantilism or AI in the chain, so this forecast establishes co-movement with the framework rather than attribution to it, exactly as forecast 13 does. That blindness is why an earlier draft cut it, and we restate the limitation rather than repair it by adding conditions we could not resolve. Second, the structural conditions are now read rather than carried, closing a gap the prior draft had to concede. USD/JPY stood at 159.41 on July 31, 2026, with the yen down 8.25% over twelve months. The Bank of Japan held at 1.0% in July after a 25 basis point hike in June, its highest policy rate since September 1995, on an 8 to 1 vote with one member pressing for more. The carry is therefore compressing from both ends, rising funding cost against a falling expected return, while one mid-year estimate still puts the differential at 250 to 275 basis points by the fourth quarter. A carry that narrows without closing is the configuration in which unwinds have historically been most violent, because positioning persists while the cushion thins. That is the setup relationship 2 describes, and it is why we hold 30% rather than the unconditional base rate.

Table 2. Resolution mechanics. First prints govern, and later revisions are ignored. Successors and renamed entities count throughout. When no substitute is specified, discontinuation of a series leaves the forecast unresolved.

#	Source of record (first print)	Threshold, ties, and fallback
1	CPCA full-year 2027 retail print; no single month resolves it	>60.0%; exactly 60.0% = NO. Fallback: CAAM wholesale, flagged as a different denominator
2	Each provider's pricing page on July 31, 2026 against December 31, 2027; standard on-demand output rates only (no batch, cached, or negotiated). Baseline reading for OpenAI: GPT-5.6 Sol at \$30 per 1M output tokens; all three readings and their sourcing are in the data files	≥50.0% drop for any one provider against its own baseline; exactly 50.0% = YES. Fallback: provider pricing announcements
3	First IEA Oil Market Report carrying a complete 2027 Chinese gasoline figure; both years read from that print	Strictly below the 2026 average; tie at reported precision = NO. Fallbacks: JODI-Oil, then S&P Global
4	MOFCOM's Chinese-language export-control list (Announcement No. 23 of 2026 and successors); point in time	Named = YES. Removal or entity-wide waiver = NO; regime-wide suspension while still listed = YES
5	First print of IFR World Robotics 2028 (published ~September 2028), China units ÷ world units	>54.0%; exactly 54.0% = NO. If the IFR discontinues the series, unresolved; MIIT data over a stale world denominator is not a substitute
6	The authority's own confirmation of a docketed investigation or action; the employee-monitoring nexus must appear in its statement of scope	Opening resolves YES permanently; later dismissal or annulment does not flip it. Preliminary inquiries and parliamentary letters do not count
7	The company's own SEC filing, official transcript, or press release; ≥1 GW across sites named in that disclosure, in the original capacity terms	Analyst or third-party reporting of lease non-renewals does not count
8	Sum of "purchases of property and equipment" cash-flow lines, four calendar quarters; Microsoft re-cut to calendar quarters; finance leases excluded	Strictly below 2027; tie at reported million = NO. Migration into JVs or vendor financing lowers the line and is accepted because that migration constitutes the retreat being forecast

#	Source of record (first print)	Threshold, ties, and fallback
9	Executed agreement disclosed by either party in a press release or securities filing; workflow data must be named in it; Fortune's sector labels at announcement govern	Pilots and letters of intent do not count
10	LMarena public overall text leaderboard, daily archived snapshots; tie at top score counts	≥30 consecutive days at #1. If LMarena is discontinued, unresolved; a different benchmark can reverse the outcome
11	Useful-life range in each company's property-and-equipment policy note against the prior year, or a disclosed change in estimate naming an equipment class	Two distinct companies; one = NO. Amazon excluded (already reduced, January 2025)
12	Census Bureau FT-900 annual totals, customs basis, goods only	Strictly below combined 2026; tie = NO. Fallback: BEA goods-imports detail. Tariff front-running likely inflated the 2026 baseline, favouring YES; noted, not adjusted for
13	BEA Personal Income and Outlays, y/y core PCE as first published for each 2027 reference month	≥6 qualifying months; exactly 3.0% qualifies. A slipped calendar does not change which print governs
14	Customs Duties net of refunds, fiscal-year-to-date column, September 2027 Monthly Treasury Statement	>\$250.0 billion. Fallback: Treasury's combined annual statement of receipts, same line
15	The company's own securities filing, official earnings-call transcript, or press release; the disclosure must name a world model or physical-AI foundation model as the source of synthetic training data and robots in its own facilities	Vendor or press case studies do not count without the operator's own statement. Pilots count only if the robots are described as operating in production. Fortune's sector labels at announcement govern
16	LA Metro press releases and board reports, first print; the 28 projects as revised by the March 2024 board action	Strictly fewer than 20 open for fare-paying passenger service on July 14, 2028; exactly 20 = NO. Groundbreakings and substantial-completion milestones do not count. Fallback: FTA quarterly project status reporting
17	Federal Reserve H.10 daily USD/JPY (FRED DEXJPUS), first print; every window of 20 consecutive H.10 trading days from August 3, 2026	$(\text{window max} - \text{window min}) \div \text{window min} \geq 10.0\%$ in any window; exactly 10.0% = YES. Fallback: Bank of Japan 17:00 JST spot, same rule

Sources and Method

Every plotted point traces to a row in `data/figure_sources.csv`, which carries the full URL, access date, vintage, transformation, and verification status for each. Claims made in prose are recorded per claim in `verification/`, including what each source does and does not establish. Principal sources by forecast:

1, 12: CPCA monthly and annual retail prints (via CnEVPost and Chinese-language coverage of the releases); State Council Information Office on 2026 purchase-tax and trade-in policy; Federation of Thai Industries (Thailand); JATO (Japan); Census Bureau FT-900. **2:** Provider pricing pages as published July 31, 2026 (baseline readings and status in `data/figure_sources.csv`; capture procedure in `data/snapshots/`); OpenAI launch announcements for the historical series. **3:** IEA *Oil Market Report*, March to May 2026, and the IEA's Strait of Hormuz supply commentary; Kpler displacement estimates; China Ministry of Public Security registrations; GL Consulting via Bloomberg. **4:** MOFCOM Announcement No. 23 of 2026 (June 22, the listing) and Announcement No. 26 of 2026 (effective July 1, the separate reporting-reward instrument). **5:** IFR *World Robotics* press releases, 2024 and 2025 editions. **6:** Regulation (EU) 2026/1744 (Digital Omnibus, in force July 27, 2026); CNIL v. Amazon France Logistique and its reduction on appeal; Hamburg DPA v. H&M (2020); Bundesarbeitsgericht 2 AZR 681/16; Meta's own MCI statements, with press reporting identified as the sole source for the corpus-exposure figures. **7, 8, 11:** Company SEC filings and earnings releases (Amazon's 2024 10-K for the useful-life reduction and the \$920 million charge; Microsoft's FY2023 10-K for the six-year extension; Alphabet and Meta extension disclosures; Meta Q2 2026 for guidance), supplemented by analyst reports of lease cancellations used solely as counterevidence. **15:** NVIDIA's Cosmos 3 technical report of June 22, 2026 (research.nvidia.com/labs/cosmos-lab/cosmos3/technical-report.pdf) and DeepMind's Genie 3 product documentation; the embodied world-model benchmark literature for the shift toward closed-loop evaluation (WorldArena 2.0, arxiv.org/pdf/2605.17912; RoboTrustBench, arxiv.org/pdf/2606.01600). The cumulative

download figure is NVIDIA's own and is cited as a vendor claim. **16:** LA Metro board reports and press releases, including the March 2024 board action that revised the project list, and the LA28 organizing committee for the opening date. **17:** Federal Reserve H.10 daily rates (FRED DEXJPUS), with the August 2024 carry unwind as the reference-class event. **13, 14:** BEA *Personal Income and Outlays* monthly first prints and the Q2 2026 advance GDP estimate; FOMC statement and implementation note, July 29, 2026; Treasury *Monthly Treasury Statements*; USTR Section 301 notice of July 23, 2026.

Method. First prints govern every resolution and plotted series. Whenever a reference class uses revised historical data, the text identifies the exception. Approximate values appear as estimates, and missing observations are never interpolated (the October 2025 PCE gap remains visible). Every plotted point corresponds to a row in `data/figure_sources.csv` containing its source, vintage, URL, access date, and verification status. `verification/` records the evidentiary basis of each prose claim. Forecast 2's July 31, 2026 baseline readings appear there as well. Direct capture of the pricing pages was blocked in this build, so those readings continue to rely on secondary reporting until the pages can be captured. This project is attached to the PDF as `ftf2026-repo.zip`: manuscript source, figure code, data, the provenance ledger, verification records, and the archived cut material.

Self-Audit

Most likely to be wrong. (1) Forecast 8: the physical constraint may be real even if the contraction arrives in 2029. The forecast scores the date, so correct mechanism and incorrect timing still produce a loss. (2) Forecast 3: post-shock price normalization could induce enough additional driving among the remaining internal-combustion fleet to exceed the growing electric fleet's displacement for a single year. That outcome would create another timing loss on the mechanism in which we have the greatest confidence. (3) Forecast 6: the enforcement base rate is thin, and the estimate extrapolates from adjacent records in the absence of direct precedent.

How correlated the set is. Seventeen forecasts amount to roughly nine independent tests once shared drivers are counted. Forecast 16 improves that ratio deliberately, since a transit program on a fixed deadline shares almost no driver with the AI buildout, while forecast 15 joins the existing diffusion cluster and forecast 17 correlates with any recession path. The delivery cluster (7, 8, 11) depends largely on whether announced capacity can be connected to the grid, and a recession would move 8, 11, and 13 together. Forecasts 1 and 3 share a single vehicle fleet; 5 and 10 deliberately measure opposite sides of one claim. Thirteen forecasts resolve within 2.7 years. A single macroeconomic path could therefore determine much of the score, which is why the falsification test evaluates patterns across resolutions.

Hardest to defend against a hostile expert. (1) The moat forecast in 9: adverse selection is theoretically persuasive but supported by little direct evidence, while functioning markets in advertising, credit, and healthcare claims emerged after brokers standardized their assets. (2) Automation follows scarcity: a development economist can point to Vietnamese and Thai factories that automate for quality control or foreign-investment requirements. Our response is that China installs most of the world's industrial robots despite wages far below US levels and a contracting working-age population. A pure labor-cost account would predict the opposite geography. (3) Europe may regulate itself out of the input market: Europe can rent frontier capability, a serious objection. Our answer depends on agentic competence being specific to an organization's processes, which remains an unsettled empirical question.

Where our evidence is weakest. Direct access to primary release archives was blocked throughout this revision. A minority of values therefore rely on reporting that quotes the primary documents. Every affected series is identified in the data files. Two gaps are material: the monthly core-PCE first prints behind Figure 4 and forecast 13, whose 2026 values we could not recheck, and the baseline reading for forecast 2. These limitations belong in the submission because they bear directly on resolution.

Conviction lines. The position we are most prepared to defend is the optionality account of China's EV and oil position, because its derivation is arithmetic and the crisis behavior is documented. Forecast 8 is the weakest: a timing bet against the most heavily financed capital-expenditure programs in history.